10 Business Questions

NYC Ride-Hailing, Weather & EV Charging Access

Course	DL3004 Data Warehousing & BI Lab, Fall 2026, FAST-NU Lahore
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These questions guide the warehouse design. Each one can be answered from our fact tables.

A. Ride-Hailing Trips

#	Business Question	Fact Table	Main Dimensions
1	Which pickup zones have the most trips each month?	Fact_Trips	Zone, Date
2	Which zones have the highest driver pay per mile?	Fact_Trips	Zone
3	How do trips, fares and tips compare between Uber and Lyft?	Fact_Trips	Company, Date
4	Which hours of the day have the longest customer waiting times?	Fact_Trips	TimeOfDay

B. Weather and Zone Demand

#	Business Question	Fact Table	Main Dimensions
5	How much does rain change the number of trips per hour?	Fact_Zone_Hourly_Snapshot	Weather, TimeOfDay
6	Does heavy rain increase customer waiting time in each borough?	Fact_Zone_Hourly_Snapshot	Weather, Zone
7	Does driver pay per mile go up when it rains?	Fact_Zone_Hourly_Snapshot	Weather, Date

C. EV Charging Access

#	Business Question	Fact Table	Main Dimensions
8	How many charging stations are in each borough, and how has this grown by year?	Fact_Charger_Coverage	Zone, Date, Station
9	Which taxi zones have no charging station at all?	Fact_Charger_Coverage	Zone
10	Which busy zones (most trips) have few or no chargers?	Fact_Zone_Hourly_Snapshot + Fact_Charger_Coverage	Zone

Note: Question 9 is answered by finding zones in Dim_Zone that have no matching row in Fact_Charger_Coverage. This is the benefit of a factless fact table.